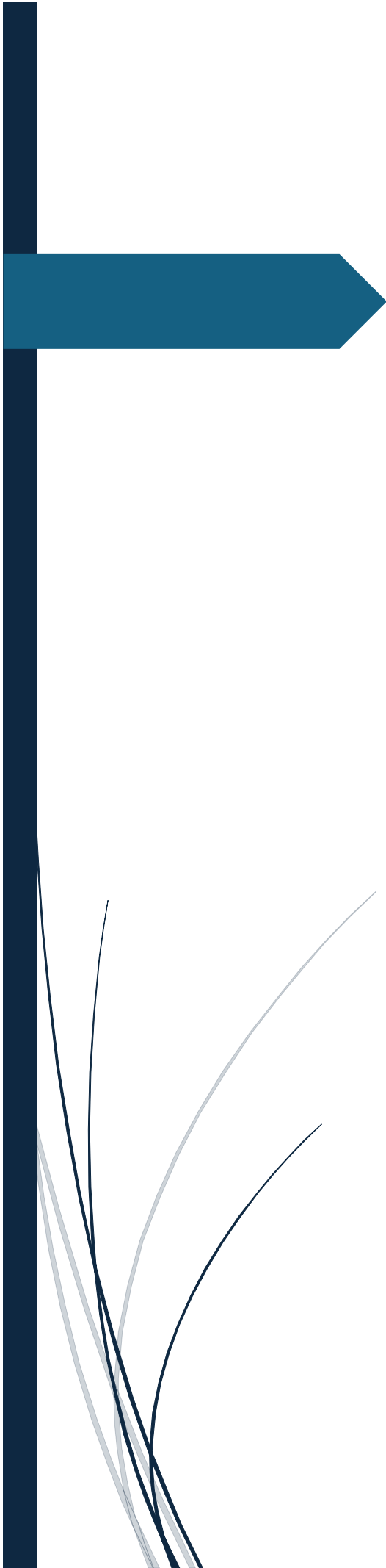




ROOFTOP SOLAR POWER INSTALLATION AT INFOPARK, KOCHI

Christy sara saji

Gisma University of Applied Sciences

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EXECUTIVE SUMMARY

This report presents a complete project management plan for a proposed rooftop solar power installation at Infopark, Kochi, structured in line with acknowledged project management frameworks and best practices. The project intends to install a grid-connected solar photovoltaic (PV) system in selected Infopark buildings, which helps to reduce energy costs, improve sustainability, and correspond with environmental, social and governance (ESG) objectives.

The projected budget is €750,000, and the duration plan is 24 months. The report is structured as a professional project management document, which covers context and objectives, planning and methodology, stakeholder management, risk management and allocation of resources, and critical evaluation of limitations. A traditional project management approach is implemented, explained by the scope, legal obligations, and fixed deliverables typical of essential system development.

1. PROJECT CONTEXT & OBJECTIVES

1.1 Project Overview

Infopark, Kochi, is a technology park owned by the Government of Kerala, which hosts more than 200 IT/ ITES companies and tens of thousands of professionals. Although Infopark is a developed organisation, it faces a critical issue of reducing its dependence on convenient grid electricity due to high energy costs and significant pressure on organisations to achieve sustainability goals. The rooftop solar power reduces Infopark's dependence on grid electricity and improves its sustainable goals.

This project report consists of the design, acquisition, installation and solar commissioning of around 1MW of grid-tied rooftop solar PV capacity in the selected buildings over a 24-month timeline, delivered in collaboration with an experienced solar EPC provider such as Tata Power Solar. The initiative seeks to:

- Minimise the reliance on the conventional grid electricity
- Optimise the operational efficiencies
- Contribute to renewable energy and decarbonisation goals

1.2 Strategic Relevance

The project is strategically relevant because it:

- Minimises the long-term electricity costs by generating solar power on site.
- Aligns with the government's renewable energy initiatives.
- Strengthens its ESG profile, which attracts investors who value sustainable practices.
- Demonstrates responsible foundational growth.

1.3 Project Objectives

The project objectives are explained using the SMART framework to make sure they are clear, measurable, achievable, relevant and time-bound.

- Based on the roof area assessment, install around 1MWp of rooftop PV capacity across 8 buildings within Infopark, Kochi, by the end of month 24.
- Reduces at least 25% of electricity consumption, which is approximately 1,500 MWh per year.
- Ensure the goal is achieved within the budget of €750,000.

- Focuses on 100% compliance with safety and environmental regulations.

1.4 Success Criteria

The project will be considered successful if it meets the following:

Criteria	Measure	Target
System commissioning	Grid connection acceptance & final approval	100%
Schedule performance	Planned vs real completion time	Within +/- 2 months
Budget	The given budget vs the final cost	Within +/- 10%
Energy targets	Predicted vs first-year generation	>= 95% of forecast
Stakeholder impression	Post-project survey	>= 8/10 average rating
Safety & compliance	No safety violations; all legal approvals in place	100%

These success criteria convert the objectives of the project into measurable goals.

2. PROJECT PLANNING & METHODOLOGY

2.1 Project Management Approach

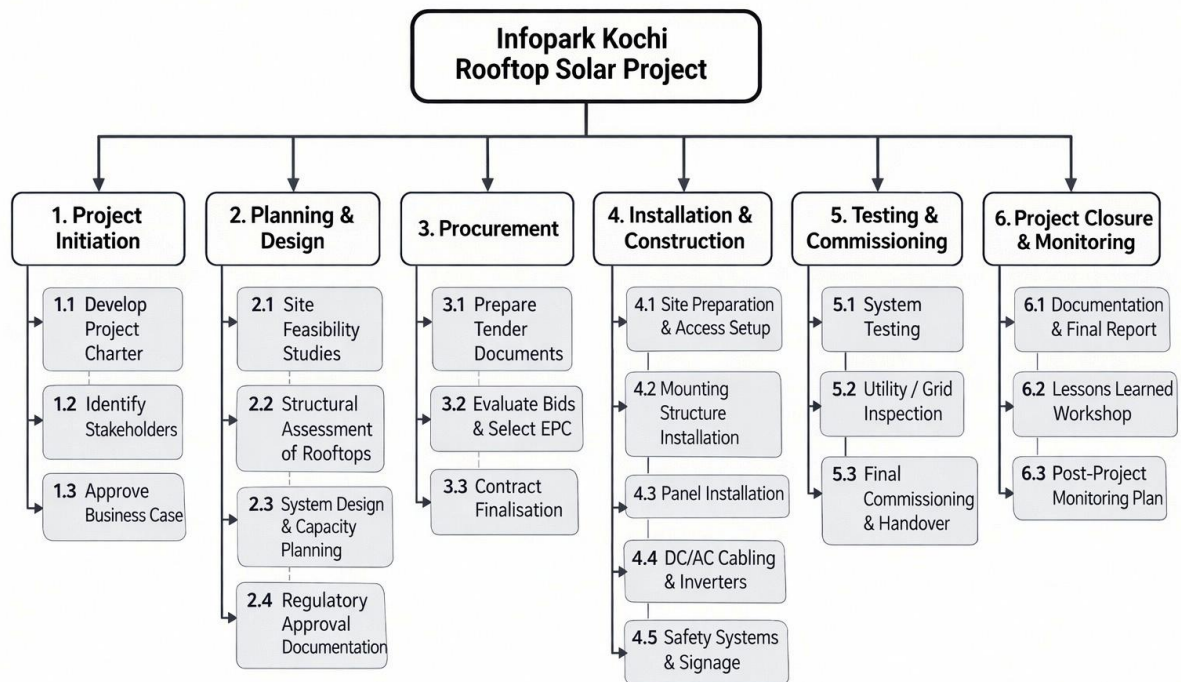
Waterfall, a traditional project management approach with stage-gate control, is adopted for this project.

Justification:

- Scope clarity: The scope and deliverables of the project are defined upfront.
- Regulatory requirements: The rooftop solar installation must satisfy safety regulations, electrical standards and other utility requirements. Thus, structured documentation should be required.
- Sequential execution: The construction and installation should follow a strict sequence. Because later steps will be completely dependent on the earlier ones.
- Cost of change: Once the installation has begun, changing the plan and resources is expensive.

Overall, this traditional approach aligns with the PMI Process Groups, such as initiation, planning, execution, monitoring and control, and closing, which are designed for methodological phase-gated delivery. (PMI,2021).

2.2 Work Breakdown Structure (WBS)



The Work Breakdown Structure (WBS) represents a top-down approach of the Infopark Kochi Rooftop Solar Project, which is divided into 6 major phases: project initiation, planning & design, procurement, installation & construction, testing & commissioning, and project closure & monitoring. Each phase is separated into controllable tasks to ensure clear scope definition, transparency, and effective control. The WBS establishes a basis for project planning, cost estimation, resource distribution, and risk identification, which could be easily monitored throughout the project lifecycle.

2.3 Project schedule & critical path

A high-level phase is scheduled:

Phase	Planned duration	Start month	End month	Notes
Initiation	2 months	1	2	Plan approval

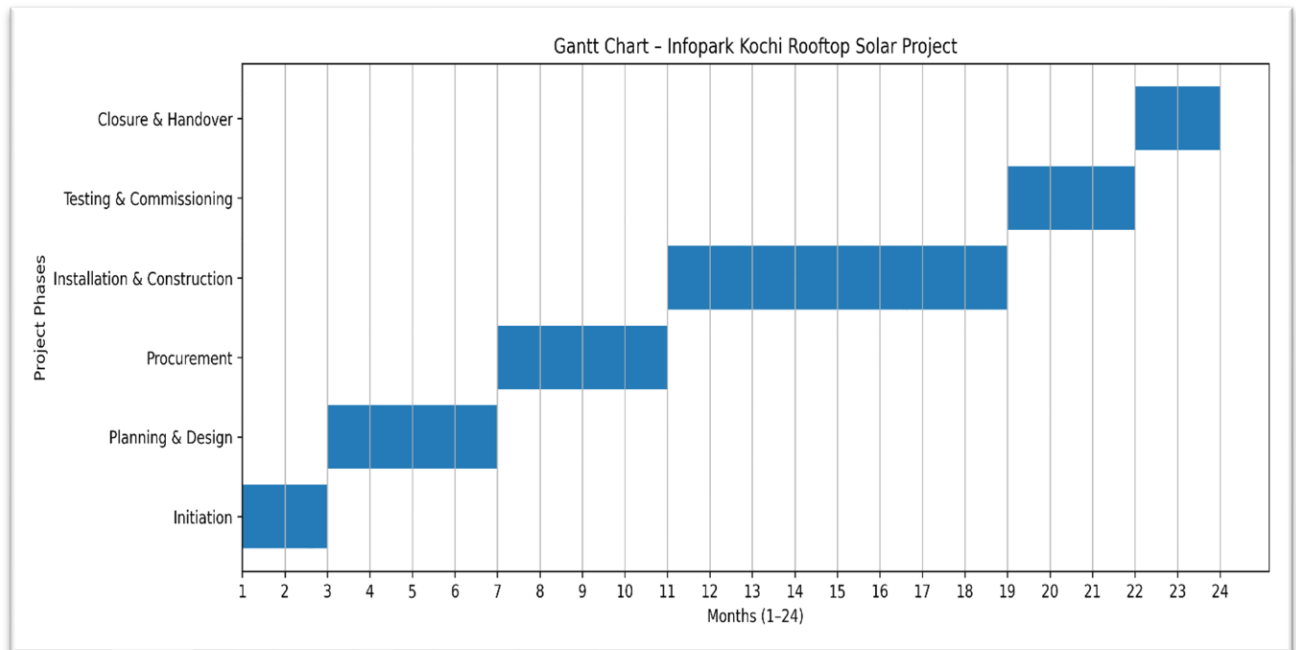
Planning & design	4 months	3	6	Finish off designs, permits applied
Procurement	4 months	7	10	Order equipment
Installation & construction	8 months	11	18	Main execution
Testing & commissioning	3 months	19	21	Grid tests & final approval
Closure	2 months	22	24	Training, O&M handover, lessons learned
Total	24 months	1	24	

The project is planned to last 24 months and is divided into six phases. Each phase has a planned duration. Thus, the implementation of each phase will be followed by the scheduled duration.

The critical path, which is the chain of tasks that cannot be delayed, includes:

- Completing designs and getting authorisation,
- Ordering and obtaining key equipment,
- Completing construction and installation,
- Testing and getting grid connections,
- Commissioning and handover.

These processes should be managed carefully to avoid any interruption.



The above Gantt chart visualises the project phase of the Infopark Kochi rooftop solar project.

3. STAKEHOLDER & COMMUNITY MANAGEMENT

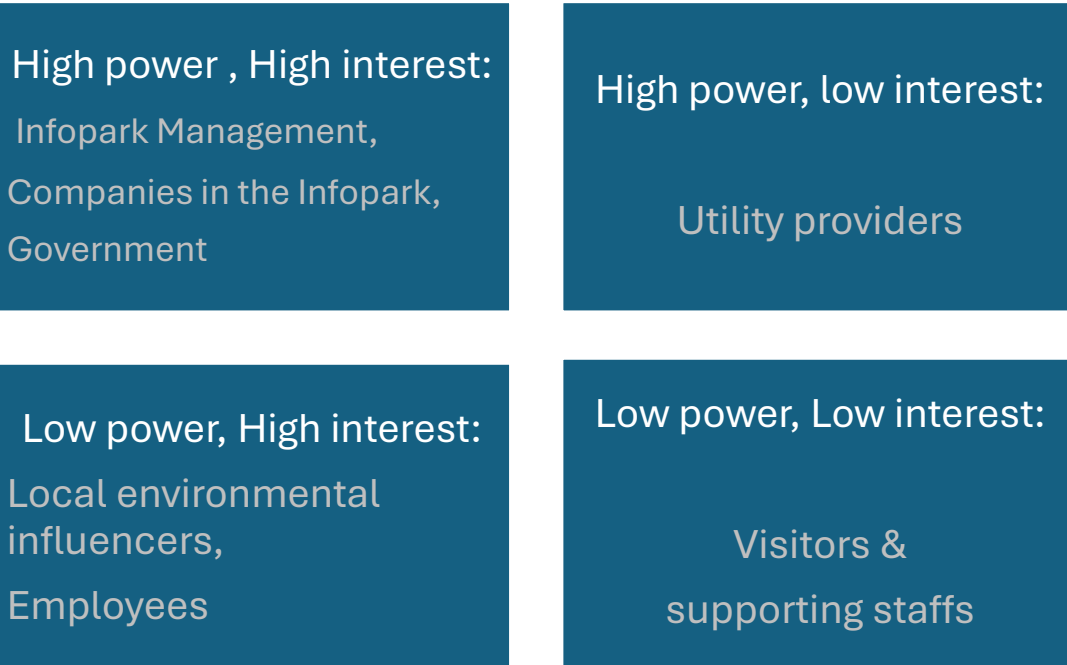
3.1 Stakeholder Identification

The key stakeholders of this project are:

- **Infopark management.** They are the decision-makers, the authorities who control the budget, executive delivery, and sustainability goals.
- **IT / BPO companies.** They are the secondary sponsor of the project, who are interested in dependable energy, cost reduction and ESG alignment.
- **Utility and grid operators.** They are the ones who must approve the solar and connect it safely to the grid.
- **Solar EPC contractor** (eg, Tata Power Solar). The delivery partners who are interested in the performance target and scope.
- **Government.** The regulatory authorities, especially Kerala's energy department and the municipal corporation.

- **Environmental communities.** Local people and influencers who care about sustainability and land impact.

3.2 Stakeholder analysis



The main stakeholders are classified into four levels using the Power-Interest Matrix based on the power and interest they have in the project. The management of Infopark, companies operating within Infopark, and the government are the top high-power, high-interest stakeholders, and their influence in managing the project will be significant throughout the project lifecycle. Utility providers have the power, but they are not very interested in monitoring throughout the years. The activists and employees in the companies at Infopark are very concerned about the project; however, the power they have is very low. Additionally, the low-power, low-interest stakeholders will be monitored with minimal commitment.

3.3 Communication strategies

The communication plan matrix is given below:

Stakeholder	Frequency	Communication Method	Owner
Sponsor of the project (CEO Infopark)	Monthly	Progress reports & committee meetings	Project manager
Infopark companies	Quarterly	Email updates	Project manager & Facility manager
EPC contractor	weekly	Technical site or virtual meetings	Project manager & Technical lead
Government regulators	As required	Formal documentation and site examination	Project manager & Compliance officer
Entire project team	weekly	Team meetings & shared project dashboard	Project manager

Communication among stakeholders is necessary for building a project. Through regular updates, all the stakeholders will be aware of the project scope, schedule, roles and risks. It also increases their commitment and support towards the solar initiative. Therefore, the timely decision-making, approval and execution occur.

4. RISK MANAGEMENT & RESOURCE ALLOCATION

4.1 Risk Identification

The main risks associated with the rooftop solar project include:

- Delays in getting authorisation for the installation and grid connection.
- Inherent problems detected through surveys, such as poor roof load capacity or water leakage.
- Weather conditions that interrupt rooftop installations.
- Budget overruns caused by the increasing cost of key materials.
- Technical issues with grid interconnection or extended testing time.
- Complaints from residents due to the operational disruptions during construction.
- Safety-related incidents during the construction and electrical installation.
- Insufficiently skilled labourers for installation and certain other tasks.
- Changes in state incentives, such as indifferent tax treatment, affect the execution of the plan.

4.2 Risk Assessment

To assess the risks identified, a probability-impact matrix is used.

Risks	Probability	Impact	Mitigation Priority
Approval delays	Medium	High	1
Structural problems	Low	High	2
Weather issues	Medium	Medium	3
Cost overruns	Medium	High	1

Technical problems	Low	Medium	4
Tenant disruption	Low	Medium	4
Safety issues	Low	High	2
Insufficiently skilled labourers	Low	High	2
Changes in government incentives	Low	Medium	4

A probability-impact matrix was used to assess the identified risks of the project. Risks such as authorisation delays and budget overruns were treated as the priority because these affect the project timeline and cost. Inherent problems and safety-related issues were treated as medium-priority because of their possible impact on workers' safety and project continuity. Risks such as technical issues, tenant disruption, etc, are treated as low priority because these are manageable by constant monitoring.

4.3 Risks Mitigation Strategies

Some of the methods that help to mitigate risks are:

- Proactive participation with the authorities. Starting permit applications and organising a meeting with regulators can speed up the approvals of the project. Assigning a compliance officer to track the permissions and escalate issues quickly.

- Carry out surveys before final designs that must be reviewed by qualified structural engineers. Keep the design choices ready so that the budget and timeline issues can be avoided.
- The construction work should be done between September and March and avoid the monsoon period. Build weather buffers and schedule the on-site work flexibly.
- Use fixed prices for the essential equipment, construction and workers. Conduct an audit check monthly against the costs and apply value engineering if any variances occur.
- Advanced technologies should be used along with reputable EPC contractors who have previous experience with similar projects. Factory acceptance test and on-site tests should be implemented properly.
- Schedule a fixed time for construction, which must be accepted by the tenants, so that the noise complaints can be avoided.
- Implement a safety plan while working at height and using electrical appliances. Provide training and work clothes to workers and conduct regular site inspections.
- Hire engineers and workers based on their skills and experiences. Introduce certain complex tasks to subcontractors.
- Track regulatory updates and policies throughout the project. Preplan the financial model with sensitivity analysis so the project remains feasible under different government incentives.

4.4 Resource Allocation

Resource allocation includes key roles, time allocation and estimated costs:

- **Project Manager (PM):** Leads overall planning, execution, maintains relationships with stakeholders and manages risks, who must be available full-time on the project (FTE 1.0). The estimated budget of the PM is €120,000.
- **Technical Lead:** Design the system, make the changes accordingly, and manage technical features with vendors, who must work full-time for more than 20 months (FTE 0.8). The cost of a solar engineer is €90,000.
- **Site Supervisor:** Leads day-to-day site activities, manages the safety of the workers and ensures quality of installation work, who must be available on the project partially over 18 months (FTE 0.6). The estimated cost of a supervisor is €40,000.
- **Electrical Engineer:** Manages grid interconnection and testing, focuses on electrical safety systems and appliances, and must work part-time for over 18 months on the project (FTE 0.5). The cost of an electrical engineer is approximately €40,000.
- **Finance Manager:** Manages overall costs, schedules procurement processes and budgets, and informs PM if any variation happens; must work part-time for 24 months completely (FTE 0.3). The estimated cost of a finance manager is around €30,000.
- **Compliance Officer:** Manages regulatory approvals and permits, ensuring the project is aligned with legal and regulatory requirements, and must work part-time on the project for at least 18 months (FTE 0.2). The scheduled budget for the compliance officer is €15,000.

Altogether, this shows the core team of the project, how much time they have to work (FTE), and roughly how much they cost the project.

4.5 Budget Overview

The given budget of the project is €750,000. The budget is allocated into:

Category	Estimated Cost
Design & consultancy	€100,000
Equipment procurement	€400,000
Installation & workers	€150,000
Permits & compliance	€30,000
Contingency	€70,000
Total	€750,000

The equipment dominates 53% of the costs due to the increasing module costs. Following that, construction and labour costs around 20%. Design and compliance share 17% with technological complexity and regulatory requirements. Contingency at 10% due to the typical construction risk exposure.

5. CRITICAL EVALUATION & REFLECTION

5.1 Limitations of the Project:

1. Fixed task durations: According to the Gantt chart, the project phase is linear, and the execution is predictable. However, the real construction depends on the weather, regulatory approvals and timely supplies. The 24-month timeline has some fundamental risks, especially in India's monsoon season and administrative processes. Though schedule buffers and pre-planning approaches reduce risk, they do not eliminate it.
2. Uncertainty in forecasting energy generation: PV output models are designed based on past weather records and technical specifications. However, the actual generation depends on seasonal

changes, and the equipment degradation is not fully predictable. ROI is sensitive to this uncertainty. A contract of the performance guarantee from suppliers is needed, and continuous monitoring is essential.

3. Traditional approach inflexibility: The Waterfall method assumes requirements are upfront. If the technology costs differ, the state incentives changes or the priorities of the tenants shifts, then the plan may become suboptimal. Thus, it is better to be steady with the plan and focus less on policy changes. Further expansion may require additional costs.
4. Stakeholder engagement: Although the stakeholder communication plan is comprehensive, actual resistance may occur during the execution. Through continuous engagement with stakeholders, build the relationship and reduce the resistance.

5.2 Lessons Learned & Reflection

The key lessons learned are:

- Bureaucratic entanglements require early engagement.
The major infrastructure projects, such as solar, power and water, are heavily monitored in India. Thus, an early meeting with the electricity board, the environment department and the municipal corporation should be conducted before project design. In regulated sectors, design and regulatory communication are vital even in traditional approaches.
- Procurement strategies for suppliers and equipment reduce risks:
PV module and inverter prices are often not constant, and supply chains for these items are widely volatile. Thus, constantly look for prices months before implementation and lock the price of the

equipment when there is a reduction. This helps in cost control and early planning and avoids last-minute hurry.

- Regulatory alignment is non-negotiable.

Safety for rooftop work, electrification, and grid connections are critical. Weak HSE culture, inexperienced workers, and inadequate insurance can cause safety-related incidents and delay the project. Thus, external audits and certification of competencies are efficient in risk mitigation.

- Stakeholder engagement improves uninterrupted execution.

The tenants and employees of the affected buildings may oppose rooftop works in the early stage. Communicating with them about the shared benefit of energy cost saving improves satisfaction and reduces disruption in the construction.

5.3 Recommendations

Some recommendations are:

- Establish a project council with Infopark CEO, EPC lead and official representatives. Monthly meetings help to remove the roadblocks and ensure project alignment with requirements.
- Execute earned value management (EVM) to manage cost performance and use the schedule variance metrics to identify the variations early.
- Implement a performance tracking system (SCADA) to track the health of the system and kWh generation.
- Organise a post-deployment review after almost a year of operation to verify ROI; collect feedback from tenants and record the lessons learned for future implementation of similar projects at other sites.

- Upon successful completion of the rooftop solar, Infopark should consider the possibility of constructing additional solar panels or a hybrid power system to boost renewable energy capacity and to reduce further energy costs.
- Sustainability key performance indicators (KPIs) such as energy efficiency and decarbonisation should be combined with other Infopark strategies to support iterative development.

6. CONCLUSION

The report has presented a complete project management plan for the implementation of rooftop solar power at Infopark, Kochi. Acting as the project manager, the report reveals how accepted project management tools and frameworks can be used in a real-life network to convey organisational and environmental benefits.

Using systematic approaches such as Work Breakdown Structure and a detailed project schedule, the scope, timeline and responsibilities of the project were clarified. The plan recognises the risk identified and other limitations, while offering mitigation strategies and recommendations. Additionally, stakeholder analysis and communication strategies ensure that engagement with stakeholders can be effective throughout the project plan.

Overall, the Infopark rooftop solar project in Kochi demonstrates the success of the complex infrastructure initiatives by using project management tools effectively. Through the execution of the project, a significant value of cost cutting, emission reduction, and ESG positioning will be delivered, while creating insights for broader sustainable practices across Kerala.

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